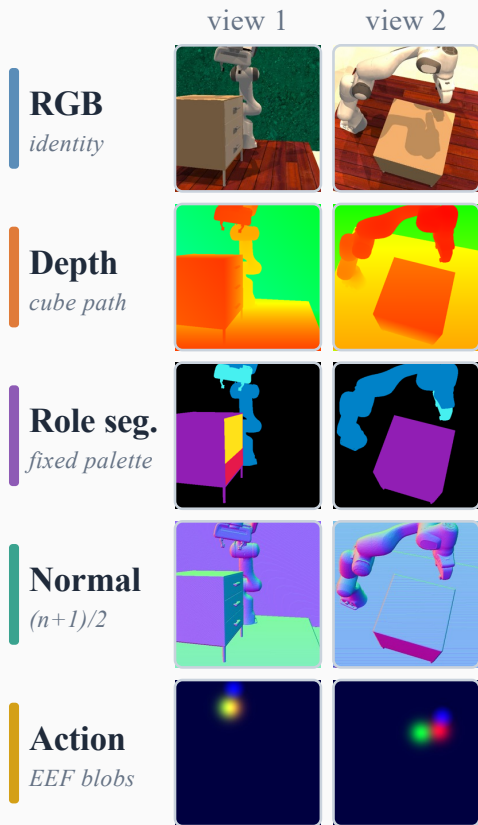


(a) Unified representation

codec γ_m : any stream $\rightarrow$ RGB video



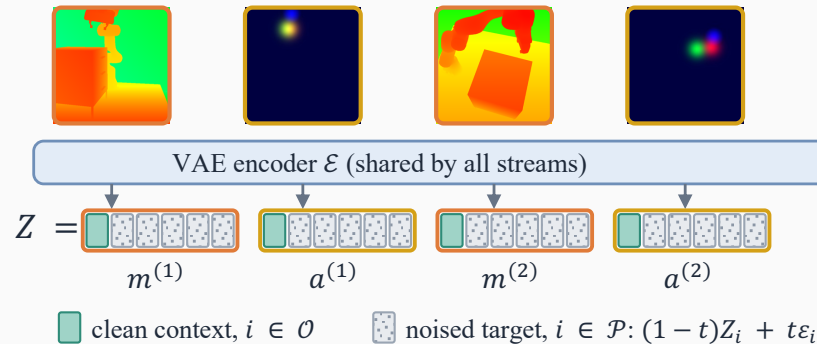
no parameters · invertible after
a frozen-VAE round trip

(b) Training: one template, one plan, one loss

① $m \sim p(\Pi)$ RGB p_1 **Depth p_2** Seg. p_3 Normal p_4 + action

② $\rho \sim p(\rho)$ **IIII joint** 1111 policy FIFT inverse

③ encode with frozen VAE $\mathcal{E}$ and pack (view-major):

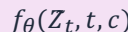


"<depth><action>
open bottom drawer"

umT5 text encoder

Shared DiT

30 blocks
one shared output layer



$f_{\theta}(Z_t, t, c)$

camera c2w, K
 $\rightarrow$ Plücker rays $\rightarrow \mathcal{E}$
 $\rightarrow$ per-block CamEnc,
added before self-attn

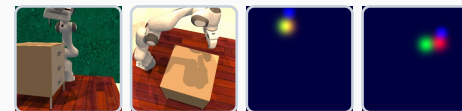
flow time $t \rightarrow$ AdaLN

$$\mathcal{L}(\theta) = w(t) \cdot (1 / |\mathcal{P}|) \sum_{i \in \mathcal{P}} \| f_{\theta}(\cdot)_i - (\varepsilon_i - Z_i) \|^2$$

One Shared Objective

(c) Inference

given: current views + current pose



<video><action> + plan II II

same weights · 50 flow steps
 $\rightarrow$ frozen VAE decoder

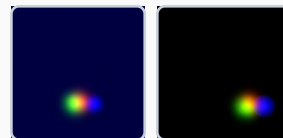
generated

frame 15 frame 29

Future RGB



Action images



γ_a^{-1} : two-view ray fusion $\rightarrow$ 7-DoF
 $\rightarrow$ IK execute $\rightarrow$ replan

cross-task generation (frame 29):



<depth> <seg> <normal>